

Pricing Logic Visual Guide

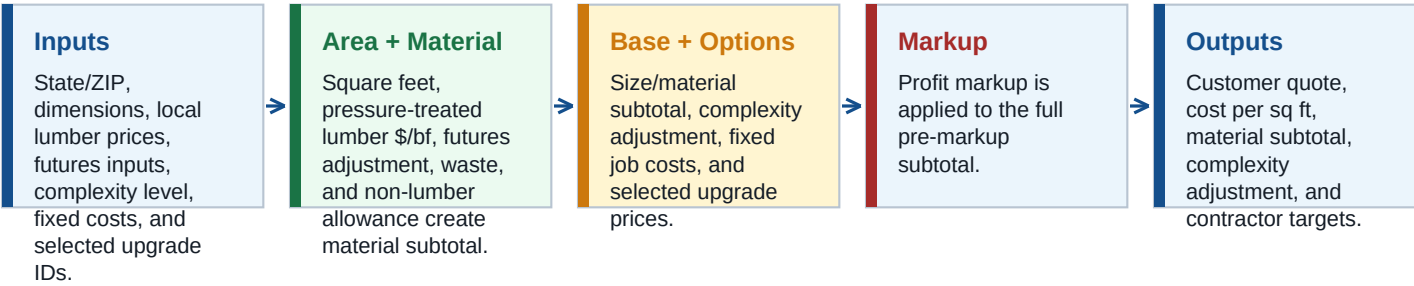
Generated from the current React calculator code on May 13, 2026. The goal is to show the live formulas, default rates, and which fields actually move price.

Catio calculator

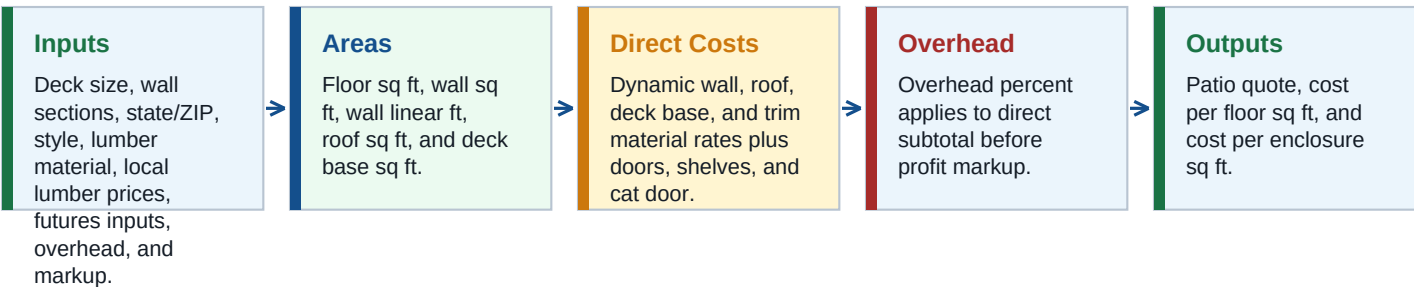
Enclosed patio calculator

Source: `src/lib/quoteCalculator.js`

Catio Price Pipeline



Enclosed Patio Price Pipeline

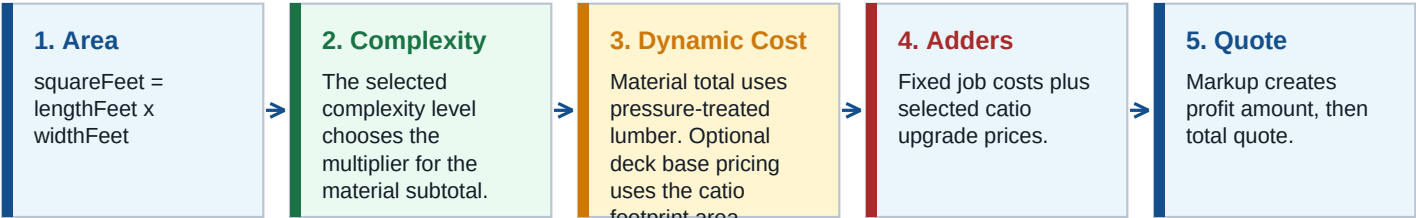


Important: some UI fields are stored for quoting context but are not currently part of the price formula. Those fields are called out later so you can decide whether they should become pricing drivers.

Calculator	Default sample output	Main pricing drivers
Catio	\$16,260.17 total, \$162.60/sq ft	State/ZIP market, length x width, local \$/board foot, futures adjustment, complexity multiplier, fixed job costs, selected upgrades, markup.
Enclosed Patio	\$18,696.81 total, \$111.29/floor sq ft	Wall sections, state/ZIP market, lumber material, local \$/board foot, futures adjustment, style multipliers, doors, shelves, overhead, markup.

Catio Formula Map

This page follows calculateQuote(). Percent values are stored as decimals, so 0.38 means 38%.



Step	Formula
Square feet	lengthFeet x widthFeet
Futures multiplier	1 + futures movement percent x futures blend percent
Adjusted pressure-treated \$/bf	local pressure-treated \$/bf x futuresMultiplier
Catio board feet	squareFeet x catioBoardFeetPerSquareFoot x waste
Catio material rate	catio non-lumber \$/sq ft + adjusted pressure-treated \$/bf x catio bf/sq ft x waste
Deck base sq ft	includeDeckBase ? squareFeet : 0
Deck base board feet	deckBaseSquareFeet x catioDeckBaseBoardFeetPerSquareFoot x waste
Deck base rate	deck non-lumber \$/sq ft + adjusted pressure-treated \$/bf x deck bf/sq ft x waste
Deck base total	deckBaseSquareFeet x catioDeckBaseMaterialRate
Selected complexity	PROJECT_COMPLEXITY_OPTIONS match by complexityLevel
Complexity multiplier	max(selectedComplexity.multiplier, 1)
Material total	catioMaterialTotal + deckBaseTotal
Complexity total	materialTotal x (complexityMultiplier - 1)
Cedar/redwood lumber upgrades	max(adjusted upgrade lumber \$/bf - adjusted pressure-treated \$/bf, 0) x totalCatioBoardFeet
Base cost total	materialTotal + complexityTotal + all fixed job costs
Catio upgrade total	sum(price) for selected catio upgrades
Quote before markup	baseCostTotal + upgradeTotal
Profit amount	quoteBeforeMarkup x profitMarkupPercent
Total quote	quoteBeforeMarkup + profitAmount
Cost per sq ft	totalQuote / squareFeet when squareFeet is above 0

Default metric	Current default value	Context
Customer price	\$16,260.17	100 sq ft with no selected upgrades.
Complexity	Standard (1x)	\$0 complexity adjustment on \$5,015.20 material subtotal.
Material rate	\$50.15	\$1.15/bf pressure-treated lumber with 448 total board feet.
Deck base default	Not included	When included, deck base uses 1.6 bf/sq ft over the catio footprint.
Before markup / markup Sky Catio quote tool pricing logic	\$11,782.73 / \$4,477.44	Markup is applied after fixed costs and selected option prices are included.

Catio Data Sources and Adjustment Notes

The base catio quote combines size/material pricing, complexity adjustment, fixed job costs, selected upgrade prices, and markup.

Fixed job costs: \$6,767.53 total

Fixed cost	Amount	Fixed cost	Amount
Flights	\$512.96	Short-term debt	\$0
Rental truck	\$774.36	Search engine optimization	\$5
Luggage	\$0	Mailchimp and Google Business Manager	\$3
Gas	\$163.18	Website services	\$40.01
Food	\$324.93	Marketing	\$218.90
Hotel	\$2,246.89	Offsite tool storage	\$701.13
Dump	\$0	New tools	\$19.48
Project manager	\$525	Unemployment insurance	\$351.34
Catio designer	\$850	Workers comp and business insurance	\$31.35

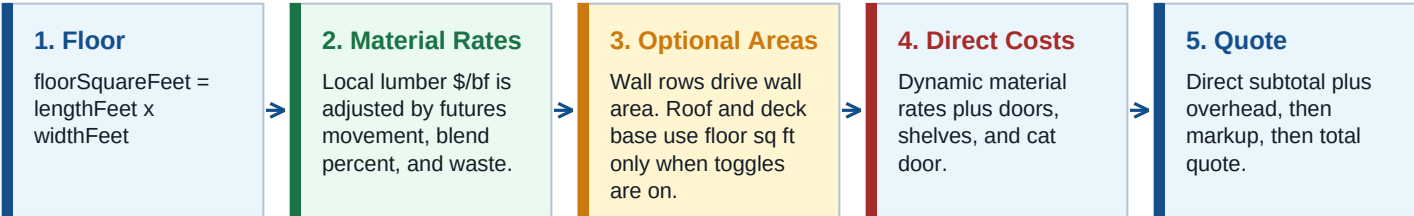
Source	Current pricing use	Adjustment lever
CATIO_UPGRADES	Selected upgrade price adds to quote. Cedar/redwood lumber upgrade prices are generated from live lumber price deltas.	Change static prices in src/data/upgrades.js or lumber assumptions in src/data/defaults.js.
Fixed job costs	Every fixed cost amount is always included when amount is above or equal to 0.	Change FIXED_JOB_COSTS in src/data/defaults.js.

Visible catio controls now feed the price formula. Deck base is priced from the catio footprint area and adds to the board feet used by cedar/redwood lumber upgrades.

Decision to make	Possible formula change
Should catio height or roof be added later?	Only add those controls when they also add wall/enclosure surface area or roof area costs.
Should deck base assumptions vary by build type?	Adjust catioDeckBaseBoardFeetPerSquareFoot or catioDeckBaseNonLumberCostPerSquareFoot in defaults.
Should selected upgrades affect complexity?	Map upgrade selections to a complexity level or add upgrade-specific complexity adjustments.

Enclosed Patio Formula Map

This page follows calculateEnclosedPatioQuote(). Wall sections are the main geometry driver.



Step	Formula	
Floor sq ft	lengthFeet x widthFeet	
Wall segment sq ft	wall.widthFeet x wall.heightFeet x wall.quantity	
Wall segment linear ft	wall.widthFeet x wall.quantity	
Enclosure sq ft	sum of wall segment square feet	
Wall linear ft	sum of wall segment linear feet	
Roof sq ft	needsRoof ? floorSquareFeet : 0	
Deck base sq ft	includeDeckBase ? floorSquareFeet : 0	
Selected lumber	LUMBER_MATERIAL_OPTIONS match by lumberMaterial	
Futures multiplier	1 + futures movement percent x futures blend percent	
Adjusted lumber \$/bf	selected local lumber \$/bf x futuresMultiplier	
Wall material rate	wall non-lumber \$/sq ft + adjusted \$/bf x wall bf/sq ft x waste	
Roof material rate	roof non-lumber \$/sq ft + adjusted \$/bf x roof bf/sq ft x waste	
Deck material rate	deck non-lumber \$/sq ft + adjusted \$/bf x deck bf/sq ft x waste	
Trim material rate	trim non-lumber \$/ft + adjusted \$/bf x trim bf/ft x waste	
Wall total	enclosureSquareFeet x wallMaterialRate x style.wallMultiplier	
Roof total	roofSquareFeet x roofMaterialRate x style.roofMultiplier	
Deck base total	deckBaseSquareFeet x deckBaseMaterialRate x style.deckBaseMultiplier	
Door total	doorCount x doorPrice	
Shelf total	shelfCount x shelfPrice	
Trim total	includeTrim ? wallLinearFeet x trimMaterialRate : 0	
Cat door total	includeCatDoor ? catDoorPrice : 0	
Direct subtotal	sum of all direct cost items	
Default metric	Current default value	Context
Patio quote	\$18,696.81	168 floor sq ft and 416 wall sq ft.
Direct subtotal / overhead	\$12,544.83 / \$1,003.59	Overhead is added before profit markup.
Cost per floor sq ft	\$111.29	Style: Screened Patio.

Enclosed Patio Defaults and Multipliers

The enclosed patio calculator prices wall enclosure area more explicitly than the patio calculator.

Patio style	Wall x	Roof x	Deck base x	Pricing effect
Screened Patio	1	1	1	Multiplies area-based costs.
Existing Porch Conversion	0.85	0.85	0.9	Multiplies area-based costs.
Premium Cat-Safe Enclosure	1.25	1.15	1.1	Multiplies area-based costs.

Default input	Value	Pricing use
Deck length x width	14 x 12	Creates floor square feet for roof/deck base and per-floor reporting.
Default walls	2x 14x8, 2x 12x8	Creates enclosure sq ft and wall linear ft.
State / ZIP market	Ohio 43215	Selects the local lumber price set that fills the board-foot prices.
Lumber material	Pressure Treated	Selects which local \$/board foot value feeds material rates.
Local lumber \$/bf	\$1.15	Anchors the dynamic material price from the state pricing lookup.
Futures current / baseline	\$575 / \$575	Creates the futures movement used to adjust local lumber.
Blend / waste	0.25 / 0.12	Controls how much futures movement matters, plus material waste.

Default direct cost item	Amount	When included
Wall enclosure	\$11,123.34	Included by count or area.
Roof area	\$0	Only when roof needed is true.
Full deck base	\$0	Only when full deck base is true.
Doors	\$650	Included by count or area.
Shelves	\$280	Included by count or area.
Wall trim	\$241.49	Only when finishing trim is true.
Cat door	\$250	Only when cat door is true.

Visible patio controls feed pricing. State and ZIP update local lumber prices; wall row width, height, and quantity drive enclosure square footage.

Adjustment Map

Use this as a quick guide when changing the pricing model or UI knobs.

What you want to change	Where to edit	Notes
Catio default rates, percents, complexity, dimensions, deck base	src/data/defaults.js	DEFAULT_QUOTE_INPUTS controls reset state and first load.
Fixed catio job cost amounts	src/data/defaults.js	FIXED_JOB_COSTS is always included in baseCostTotal.
Catio upgrade prices	src/data/upgrades.js	Selected price values are added before markup.
Catio core formulas	src/lib/quoteCalculator.js	calculateQuote() owns the pricing math and returned summary values.
Patio material rates, toggles, wall rows	src/data/defaults.js	DEFAULT_ENCLOSED_PATIO_INPUTS controls reset state and first load.
Patio style multipliers	src/data/defaults.js	ENCLOSED_PATIO_STYLE_OPTIONS controls wall, roof, and deck multipliers.
Patio core formulas	src/lib/quoteCalculator.js	calculateEnclosedPatioQuote() owns the pricing math.
Visible fields and labels	src/components/*.jsx	QuoteForm and EnclosedPatioCalculator render input controls.
Numeric input parsing	src/App.jsx	Add new numeric fields to numericFields so values are stored as numbers.

High-leverage pricing decisions to confirm

Topic	Why it matters
Catio uses footprint size	Length and width drive the catio material subtotal, and the deck base toggle uses that same footprint area.
Upgrade price is customer-facing	Upgrade price adds to the quote as a direct option adder before markup.
Patio walls are explicit	Patio pricing is more geometry-based; changing wall rows immediately changes enclosure sq ft and trim linear ft.
Markup order differs from overhead	For patio, overhead is added first, then markup. For catio, there is no separate overhead step.

Regenerate this PDF after formula or data changes by running: node scripts/generate-pricing-logic-pdf.mjs